Information Theory Mn908 and MICAS

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Exercise 3 of 7 October 2026

(solutions are due on 14 October 2026, 8h30)

Problem 1 ***On Fano's Inequality***

Let X take value in the set $\mathcal{X} = \{1, \ldots, m\}$, and let $p_i = \Pr[X = i]$ for $i = 1, \ldots, m$. Assume $p_1 \geq p_2 \geq \cdots \geq p_m$.

1. Suppose that you must guess X. Which guess would have the smallest probability of error? What is the probability of error P_e^* associated with this guess?
2. Maximize $H(X)$ over all choices of $p_1, p_2, \ldots, p_m$ for which the best guess has a probability of error P_e^*.

Problem 2 ***Markov Process and Entropy Rate***

Let $\{X_k\}_{k=1}^{\infty}$ be the following homogeneous (time-invariant) Markov process. For each positive integer $k > 1$, if $X_{k-1} = 0$, then $X_k = 0$ with probability $1/3$ and $X_k = 1$ with probability $2/3$, and if $X_{k-1} = 1$, then $X_k = 0$ with probability $1/2$ and $X_k = 1$ also with probability $1/2$.

a) Find the entropy rate $H(\{X_k\})$.

b) For each positive integer k, define $V_k = X_{2k}$. Is the process $\{V_k\}$ Markov? Be sure to justify your answer!

c) Which of the following two statements is true: $H(\{V_k\}) \geq H(\{X_k\})$ or $H(\{V_k\}) \leq H(\{X_k\})$?

d) For each positive integer k, define $Y_k = X_{k+1}$ if k is odd and $Y_k = X_{k-1}$ if k is even. Is the process $\{Y_k\}$ a Markov process? Be sure to justify your answer!

e) Find the entropy rate $H(\{Y_k\})$.

Problem 3 ***Window-Based Lempel-Ziv***

1. Encode the following sequences with the window-based Lempel-Ziv algorithm seen in class with window size 4 and using the simpler version of the unbounded integer encoding to encode the parameter L:
   - `1100000001111100111111110000`
   - `00000111011100111`

   What is your compression rate?
2. Decode the following sequence that has been encoded with the window-based Lempel-Ziv algorithm with window-size 8:
   - `0110001111101000000110011111010000`